

**Question:**  
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A.  $2.48 \times 10^{-8}$  cm
  - B.  $5.92 \times 10^7$  cm/sec
  - C.  $8.66 \times 10^{-6}$  cm
  - D.  $3.14 \times 10^{-5}$  cm
  - E.  $9.81 \times 10^{-4}$  cm
  - F.  $5.27 \times 10^{-9}$  m
  - G.  $1.23 \times 10^{-7}$  cm
  - H.  $4.14 \times 10^{-10}$  m
  - I.  $6.625 \times 10^{-27}$  erg sec
  - J.  $0.512 \times 10^6$ eV

**Reasoning + Answer:**

<tool\_call> reasoning process here

To find the de Broglie wavelength of an electron given its kinetic energy , we use the de Broglie wavelength formula :

$\lambda = h / p$

- where :
- h = Planck's constant =  $6.626 \times 10^{-34}$  J $\cdot$ s (or  $6.626 \times 10^{-27}$  erg  $\cdot$  s),
  - p = momentum of the electron .

For a non-relativistic electron , kinetic energy (K.E.) =  $(1/2)mv^2$  , and the momentum  $p = mv$  .

So ,  $p = \sqrt{2mK.E.}$

Therefore , de Broglie wavelength :

$\lambda = h / \sqrt{2mK.E.}$

We need to compute this with  $K.E. = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$  .

Mass of electron ,  $m = 9.109 \times 10^{-31} \text{ kg}$

Step-by-step :

- 1 . Compute  $2mK.E.$  :  
 $2 \times (9.109 \times 10^{-31}) \times (1.602 \times 10^{-19}) = 2.917 \times 10^{-49} \text{ kg} \cdot \text{J}$
- 2 .  $\sqrt{2mK.E.} = \sqrt{2.917 \times 10^{-49}} \approx 1.708 \times 10^{-25} \text{ kg} \cdot \text{m/s}$
- 3 .  $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$
- 4 .  $\lambda = 6.626 \times 10^{-34} / 1.708 \times 10^{-25} \approx 3.88 \times 10^{-9} \text{ m}$

Convert to cm :  $3.88 \times 10^{-9} \text{ m} = 3.88 \times 10^{-7} \text{ cm}$

Wait ⚠ let's recheck units .

Actually ,  $1 \text{ m} = 100 \text{ cm}$  ⚡ so  $3.88 \times 10^{-9} \text{ m} = 3.88 \times 10^{-7} \text{ cm}$

But look at the options : the choices are in cm , but one is given in m .

Let ⚠s convert to cm :

$3.88 \times 10^{-9} \text{ m} = 3.88 \times 10^{-7} \text{ cm}$

Wait ⚠ that's not matching any of the options .

Wait ⚠  $3.88 \times 10^{-9} \text{ m}$  is  $3.88 \times 10^{-7} \text{ cm}$ ? Yes .

Check if option F is in meters :  $5.27 \times 10^{-9} \text{ m}$  ⚡ is that close?

Wait , let ⚠s double-check the calculation .

Use standard formula for electron wavelength in meters :

$\lambda \text{ (in m)} = \sqrt{2 \times 9.11 \times 10^{-31} \times 1.602 \times 10^{-19}} / (6.626 \times 10^{-34})$

Wait better use standard shortcut :

$\lambda = h / \sqrt{2mE}$

Compute numerically :

$m = 9.11 \text{e-31 kg}$   
 $E = 1.602 \text{e-19 J}$

So  $p = \sqrt{2 \times 9.11 \text{e-31} \times 1.602 \text{e-19}} = \sqrt{2.917 \text{e-49}} = 5.401 \text{e-25 kg} \cdot \text{m/s}$

$h = 6.626 \text{e-34}$

$\lambda = 6.626 \text{e-34} / 5.401 \text{e-25} = 1.227 \text{e-9 m}$